

Nitya Naman

Engineering imagination, one pixel at a time.

<https://ciderboi.xyz> | <https://github.com/namanyt> | <https://linkedin.com/in/nityanaman> | me@ciderboi.xyz

Education

University of Petroleum and Energy Studies

B.Tech. in Computer Science and Engineering (Graphics & Gaming)
CGPA: 8.50

Dehradun, India

2025 - 2029

Experience

Full-Stack Developer

Government of Uttarakhand

- Architected and deployed the Official web platform for the 28th All India Forest Sports Meet (AIFSM 2025), supporting 3,390+ participants across 42 teams nationwide.
- Built end-to-end event management functionality including participant registration, authentication, event administration, result publication and organizer workflow
- Owned 100% of frontend, backend, database, deployment, and production infrastructure development throughout the project lifecycle

Dehradun, India

Oct 2025 – Nov 2025

Head of Club (Techformers' Club)

Delhi Public School

- Led technical operations, media production, and event logistics for multiple school-wide initiatives, coordinating photography, videography, livestreaming, and AV systems.
- Designed and operated gaming and livestreaming infrastructure for school esports competitions and mentored students for inter-school programming competitions
- Mentored 20+ junior students for inter-house creative competitions, contributing to multiple podium finishes

Jaipur, India

Apr 2023 – Mar 2025

Skills

Languages:	TypeScript, Python, Java, C, C++, C#, Rust
Frameworks:	React, Next.js, Node.js, Express, Dioxus
Databases:	PostgreSQL, MySQL, MongoDB, Prisma ORM
Infrastructure:	Linux, Docker, Nginx, SMTP, Grafana, n8n, PM2
Game Dev:	Unity, OpenGL, SFML

Projects

300 Billion Digit Computation of $\Gamma(1/5)$

High Performance Computing / Numerical Analysis (2026)

- Computed $\Gamma(1/5)$ to 300 billion decimal digits using y-cruncher on a dual-socket Intel Xeon Platinum HPC cluster with 300 GiB RAM and ~1.8 TiB external-memory storage.
- Verified the computation using two independent algorithms (Vidūnas–Lamprecht (2024) and Incomplete Gamma), producing identical hexadecimal and decimal outputs.
- Managed multi-week checkpointed computations and optimized memory, storage, and execution for large-scale arbitrary-precision arithmetic.
- Officially recognized by the y-cruncher for the largest verified computation of $\Gamma(1/5)$ to 300 billion decimal digits.
- <https://www.numberworld.org/y-cruncher/>

Student Project Management Portal (SPM)

Full-Stack Academic Evaluation Platform (2026)

- Designed a highly relational PostgreSQL database acting as a deterministic state machine to handle complex, multi-role RBAC workflows and historical rubric snapshots.
- Built the infrastructure to support 3,000+ concurrent students and faculty evaluations.
- Implemented a granular, immutable system-wide audit logging subsystem paired with automated data pipelines to ingest and normalize fragmented institutional data.

Homelab Infrastructure

Self-hosted servers (2019)

- Built a multi-node bare-metal private cluster managing 30+ containerized services—including automated n8n pipelines, Minio object storage, relational databases, localized Synapse matrix server and a smtp server.
- Engineered full edge-routing via Nginx reverse proxy across Ubuntu Linux environments, maintaining resource allocation and monitoring across Ubuntu Linux environments with 400+ days of continuous uptime (and still going).
- <https://ciderboi.xyz/brain/setups/homelab>

Engine

Learning / Graphics & Engine Framework (2026)

- Built a reusable C++ game engine framework with a modular OpenGL renderer, ECS architecture, runtime systems, and asset pipeline.
- Designed extensible engine architecture through engine–application separation, registration-based runtime management, and public APIs.
- Implemented atmospheric rendering, HDR post-processing, scene management, and integrated debugging/profiling tools.
- <https://github.com/namanyt/engine>

Fython

Learning / Finance Platform (2026)

- Developed a personal finance management platform featuring transaction ingestion pipelines, automated categorization, anomaly detection, terminal-based analytics dashboard, and desktop GUI support.
- <https://github.com/namanyt/fython>

tictactoe.c

Learning / Technical Platform (2025)

- Built a modular game in C featuring minimax AI, alpha-beta pruning, persistent player statistics, leaderboards, and multi-mode gameplay architecture.
- <https://github.com/namanyt/tictactoe.c>

todo.rs

Learning / Todo app (2025)

- Developed a native Rust desktop application with local persistence, file-based storage, and GUI workflows to explore Rust application development.
- <https://github.com/namanyt/todo.rs>

Awards / Award Winning Projects

Koding Books

Educational / Technical Platform (2023)

- Built an online technical resource platform containing 1,000+ curated programming and technology resources with categorization, indexing, and retrieval systems.
- Designed scalable content structures to improve discoverability and support future expansion.
- Received 1st place in a national-level competition for the project.
- <https://capitall.ciderboi.xyz>

1st Place

GPSC Capitall 2023

Lab of Shadows (LOS)

Educational Serious Game (2024)

- Developed a horror-themed educational science game in Unity (C#), engineering modular state machines for player mechanics and dynamic inventory tracking.
- Designed gameplay systems that mapped scientific concepts directly onto progression mechanics to drive engagement.
- Optimized asset delivery and rendering pipelines for stable performance on low-spec hardware.
- Received 3rd place in a national-level competition for the project.
- <https://ciderboi.xyz/extras/los.pdf>

3rd Place

GPSC Capax Infinium
2024

SpineSync

Wearable Posture Correction System (2024)

- Designed and developed a wearable posture-monitoring prototype using Arduino Nano and flex sensors to detect unhealthy spinal positioning.
- Built a companion Android application in Flutter for monitoring posture data and user feedback workflows.
- Engineered both hardware and software components independently, integrating embedded sensing with mobile application interfaces.
- Received 3rd Place in a national innovation competition, evaluated across both technical implementation and business viability.

3rd Place

All India Inter-DPS
Science and Commerce
Festival